Mahesh Jivaji Shriram

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Professional Summary

AI Engineer with experience building production-grade **LLM-powered applications, Agentic AI systems, and retrieval-augmented solutions**. Experienced in designing **multi-agent workflows, AI-assisted engineering systems, MCP integrations and AI automation pipelines** for enterprise software engineering use cases. Hands-on exposure to **LLM fine-tuning, knowledge distillation, model quantization, embeddings, semantic search, and model evaluation**, with a strong foundation in **Machine Learning, NLP, PyTorch, TensorFlow and Python**. Proven experience taking AI solutions from experimentation to production with measurable engineering impact.

Technical Skills

- **Programming:** Python, C++, Javascript
- **Machine Learning:** Scikit-learn, TensorFlow, PyTorch, NLP, Feature Engineering, Hyperparameter Tuning, Model Evaluation, MLflow
- **LLM Engineering:** HuggingFace, Transformers, SFT, LoRA, QLoRA, Knowledge Distillation, Response Distillation, Logit Distillation, Model Quantization, Prompt Engineering
- **Agentic AI:** LangChain, LangGraph, AI Agents, Multi-Agent Systems, RAG, Context Engineering, MCP, Agentic Workflows, Human-in-the-Loop (HITL), Guardrails
- **Evaluation & Experimentation:** RAGAS, DeepEval, Weights & Biases, LLM Benchmarking, Hallucination Analysis, Quality, Latency & Memory Evaluation
- **Backend & Deployment:** FastAPI, REST APIs, PostgreSQL, pgvector, MySQL, GitHub Actions, CI/CD, Firebase, Vector Databases

Experience

Harbinger Group — July 2025 – Present

AI Engineer

AI-Assisted Engineering Initiative

- Designed and implemented **AI-assisted engineering workflows** for code generation, auditing, validation, and repository integration across enterprise software development teams.
- Built agentic workflows integrating **LLMs, MCP tools, GitHub, AI Hooks, and CI/CD pipelines** to automate engineering tasks and improve development efficiency.
- Implemented **spec-driven workflows and pre/post-tool guardrails** to improve the reliability and consistency of AI-generated changes.
- Saved **1,875+ engineering hours across 238 AI-assisted tasks**, achieving **90.8% backend accuracy** and **88.6% frontend accuracy**.

Bug-Fixing Agent – AI-Powered Defect Resolution System

- Designed and built a **multi-agent AI system** for defect investigation, root-cause analysis, fix generation, and validation across **.NET and Angular** applications.
- Integrated **Atlassian and GitHub MCP** with CodeGraph for contextual code understanding and impact analysis.
- Implemented **HITL, browser-assisted validation, and guardrails** for reliable and controlled auto-

mated remediation.
- Achieved **84% success across 42 defects**, saving **207+ engineering hours**.

DeepPulse - AI-Based Employee Feedback System

- Built an **LLM-powered feedback system** using **RAG, LangChain, and vector search** to generate personalized employee insights.
- Developed **context-aware AI workflows** combining employee data, retrieved knowledge, and organizational objectives.
- Implemented an end-to-end pipeline for **data ingestion, embeddings, retrieval, context construction, and response generation** using PostgreSQL/pgvector.

Hartaraf Nov 2024 – Feb 2025

Software Engineer Intern (Remote)

- Developed an **intent classification and NLP pipeline** for natural-language service discovery on a Flutter-based platform.
- Implemented **Speech-to-Text (STT) and Text-to-Speech (TTS)** pipelines for voice-based interactions.
- Integrated **RAG-based service matching** with Firebase and REST APIs to retrieve contextually relevant local service providers.

Independent Projects

ModelForge - LLM Fine-Tuning, Distillation & Quantization

- Built a **model adaptation pipeline** for experimenting with **SFT, LoRA/QLoRA, knowledge distillation, and model quantization** on compact instruction-tuned LLMs.
- Implemented **response and logit distillation** to transfer capabilities from larger models to smaller models while preserving task performance.
- Compared **base, fine-tuned, distilled, and quantized models** across quality, inference latency, and memory footprint.
- Analyzed the trade-offs between **model quality, efficiency, and resource requirements** for deploying LLMs in resource-constrained environments.

ModelLens - LLM Evaluation & Benchmarking Framework

- Built an **automated evaluation framework** using **RAGAS and DeepEval** to measure faithfulness, answer relevance, context precision, and hallucination-related failures.
- Designed **domain-specific evaluation datasets and benchmarking workflows** to compare base, fine-tuned, and quantized models across quality and inference performance.
- Integrated **Weights & Biases (W&B)** for experiment tracking, metric analysis, and regression detection across model iterations.

Education

Bachelor of Engineering in Artificial Intelligence and Data Science

PES Modern College of Engineering, Pune August 2021 – June 2025

Cumulative GPA: 8.2